UMHack 2026 — Foreign Workforce Compliance

PermitIQ

AI-Powered Autonomous Compliance Engine

for Malaysian Foreign Worker Management

Compliance

Agentic AI

Regulatory

Verified

🚀 Agentic AI | LangGraph • FastAPI • Firebase | 🏆 UMHack 2026

THE PROBLEM

Malaysian SMEs face a survival-grade compliance crisis in managing foreign workers.

→ Regulations span 3+ government agencies (JTKSM, KDN, JIM) with separate portals

→ A single expired permit = **RM 10,000–50,000 fines** per worker + imprisonment

→ SMEs rely on spreadsheets — no system connects permits, medicals, levies

→ 13MP mandates reducing foreign workers to **10% by 2030**

→ June 2026 EP salary thresholds **double** — no grace period

RM 50K

Max fine per worker (Sec 55B)

3 Gates

Agencies to navigate per worker

30 Days

FOMEMA deadline after arrival

2× Levy

MTLM jump for high-dependency

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WHY NOW?

2026 is the tipping point — the regulatory environment has fundamentally changed.

⚖️ 13th Malaysia Plan (13MP)

- 10% foreign workforce cap by 2030
- Multi-Tier Levy penalizes high-dependency
- AI-first compliance (National AI Action Plan)

🚨 June 2026 EP Salary Pivot

- EP Cat I: RM 10K → RM 20K
- EP Cat II: RM 5K → RM 10K
- EP Cat III: RM 3K → RM 5K
- No grace period for renewals

📋 Strict Liability Doctrine

- Oversight = criminal offense
- 5+ undocumented = mandatory prison
- Unprecedented cost of ignorance

🧩 Knowledge Barrier

- Complexity of 3-Gate Architecture
- Strict documents & deadlines
- One gate failure = system collapse

THE COMPLIANCE JOURNEY

Every foreign worker must pass through 3 government gates + post-arrival compliance.

Gate 1: JTKSM
Sec 60K
Employer Ready

Gate 2: KDN
Quota Allocation

Gate 3: JIM
VDR Visa
FWCMS

Transit
Worker Arrival

FOMEMA
Medical 30
Days

PLKS + i-Kad
Permit Issued

Each gate is managed by a different government agency with separate portals, documents, and timelines.

Failure at any stage = entire application collapses.

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WORKER COMPLIANCE

What a foreign worker needs to prepare for compliance.

📄 Pre-Arrival (VDR Stage)

- Valid passport — min 18 months validity
- Biometric-compliant photo — white bg, no glasses, 35×50mm
- Bio-medical reference number — pre-screening in home country
- Signed employment contract with registered Malaysian company
- Academic certificates (if Employment Pass)
- Succession plan (if EP Category II/III)

🏥 Post-Arrival (PLKS Stage)

- MDAC — Malaysia Digital Arrival Card verified Day 1
- SEV stamp — Immigration entry stamp verification
- FOMEMA registration — must register within 7 days
- FOMEMA medical exam — pass within 30 days
- Biometric capture — fingerprints at Immigration
- i-Kad issuance — digital identity card

⚠ If FOMEMA Result = "Unfit"

Cat 1 communicable disease (TB, HIV, Hep B) → immediate repatriation. NCD (Diabetes) → 6-month monitoring option.

EMPLOYER COMPLIANCE

What a company must prepare to hire foreign workers.

🏢 Gate 1: JTKSM

- SSM registration — active standing
- Act 446 certificate — housing
- Section 60K approval
- Housing inspection report

📋 Gate 2: KDN

- MYFutureJobs — 30+ days
- MTLM ratio thresholds
- Sector-specific quota
- Levy payment capability

💰 Ongoing Duties

- Tiered levy payments
- Permit renewals (T-90/60/30)
- FOMEMA annual re-screening
- EP salary compliance

🔴 The Reality Check

An SME with 50 foreign workers must track:

- 150+ expiry dates
- 50 annual screenings
- 3 quota allocations

This is impossible with spreadsheets.

OUR SOLUTION

PermitIQ — An autonomous agentic compliance engine for the 2026 regulatory era.

Replaces fragmented spreadsheets with AI that reasons about regulations, tracks every deadline, and executes compliance actions.

Multimodal Document Triage

Upload passports, clinic notes in English, Malay, Bengali

Agentic AI Reasoning

LangGraph Auditor, Strategist & Filing agents

Kanban Gate Pipeline

Workers flow through gates automatically in real-time

What-If Levy Simulator

Model MTLM & salary threshold impacts

Deadlock & Expiry Detection

T-90/60/30 alerts via continuous compliance sweep

Human-in-the-Loop Safety

High-stakes decisions require explicit human approval

🚀 Agentic AI | Regulatory Compliance Engine | 🏆 UMHack 2026

HOW IT WORKS

From document chaos to autonomous compliance — in one pipeline.

📄 Bulk Upload → 🤖 AI Triage → 🔍 Audit Agent → 📊 Strategist → 📝 Filing Agent → ✅ Action Plan

🧠 The Agentic Brain

→ Auditor Agent — validates passport, photo biometrics, salary thresholds

→ Strategist Agent — calculates MTLM levy, models EP salary pivots

→ Filing Agent — auto-generates IMM.47, justification letters, COM docs

📋 Automated Workflows

→ VDR Application Manager — checklist-driven auto-validation

→ PLKS Post-Arrival Tracker — countdown & Day 7/25/30 alerts

→ Compliance Sweep — daily Celery job scans all workers

TECH STACK

Built on a production-grade agentic architecture.

Layer	Technology	Role
Intelligence	**Gemini (GLM)**	Multimodal parsing, agent reasoning, legal analysis
Orchestration	**LangGraph**	Cyclic state graphs, agent coordination, HITL interrupts
Backend	**FastAPI**	Async REST APIs, WebSocket real-time updates
Task Queue	**Celery + Redis**	Background: parsing, sweeps, FOMEMA monitoring
Database	**Cloud Firestore**	Real-time worker & compliance state tracking
Storage	**Firebase Storage**	Secure document storage (passports, contracts)
Frontend	**React 18 + Vite**	Dashboards, React Flow graphs, Kanban pipeline
Observability	**Arize Phoenix**	Full agent traceability — every step auditable

PROBLEM → SOLUTION MAPPING

Every pain point has a direct PermitIQ answer.

Problem	PermitIQ Solution	Impact
3 agencies, no connection	**Unified Kanban gate pipeline**	Single dashboard
Expired permits unnoticed	**Compliance sweep (T-90/60/30)**	Zero missed deadlines
Manual document extraction	**Multimodal AI triage (3-lang)**	Minutes, not hours
MTLM levy cost surprises	**What-If simulator**	Budget with confidence
June 2026 EP blindspot	**Strategist pre-scan renewals**	Months of warning
FOMEMA 30-day misses	**PLKS tracker + auto-escalation**	Day 7/25/30 alerts
Decisions without evidence	**HITL with AI reasoning + audit**	Defensible record

IMPACT

Measurable outcomes for Malaysian SMEs.

RM 0
Compliance fines with proactive sweep

90%+
Reduction in manual tracking effort

<3 min
Document ingestion per worker

100%
Deadline coverage across all gates

🇲🇾 National Alignment

→ Ekonomi MADANI — Digital compliance for SME resilience
→ 13MP — Supports foreign workforce reduction targets
→ National AI Action Plan 2030 — Agentic AI standard
→ MyDIGITAL Blueprint — Digital access for all businesses

🎯 Target Users

→ SME employers managing 10-500 foreign workers
→ HR managers tracking permits, medicals, levies
→ Labour consultants with multi-client portfolios
→ F&B operators with dual federal + municipal compliance

CLOSING

The era of manual compliance is over.

PermitIQ transforms foreign worker management from a reactive spreadsheet nightmare into an autonomous, AI-driven system that reasons about regulations, tracks every deadline, and shields SMEs from catastrophic fines — all before a single permit expires.

The 13th Malaysia Plan demands a new standard.

PermitIQ is that standard.

🛡️ **PermitIQ — Autonomous compliance for 2026**